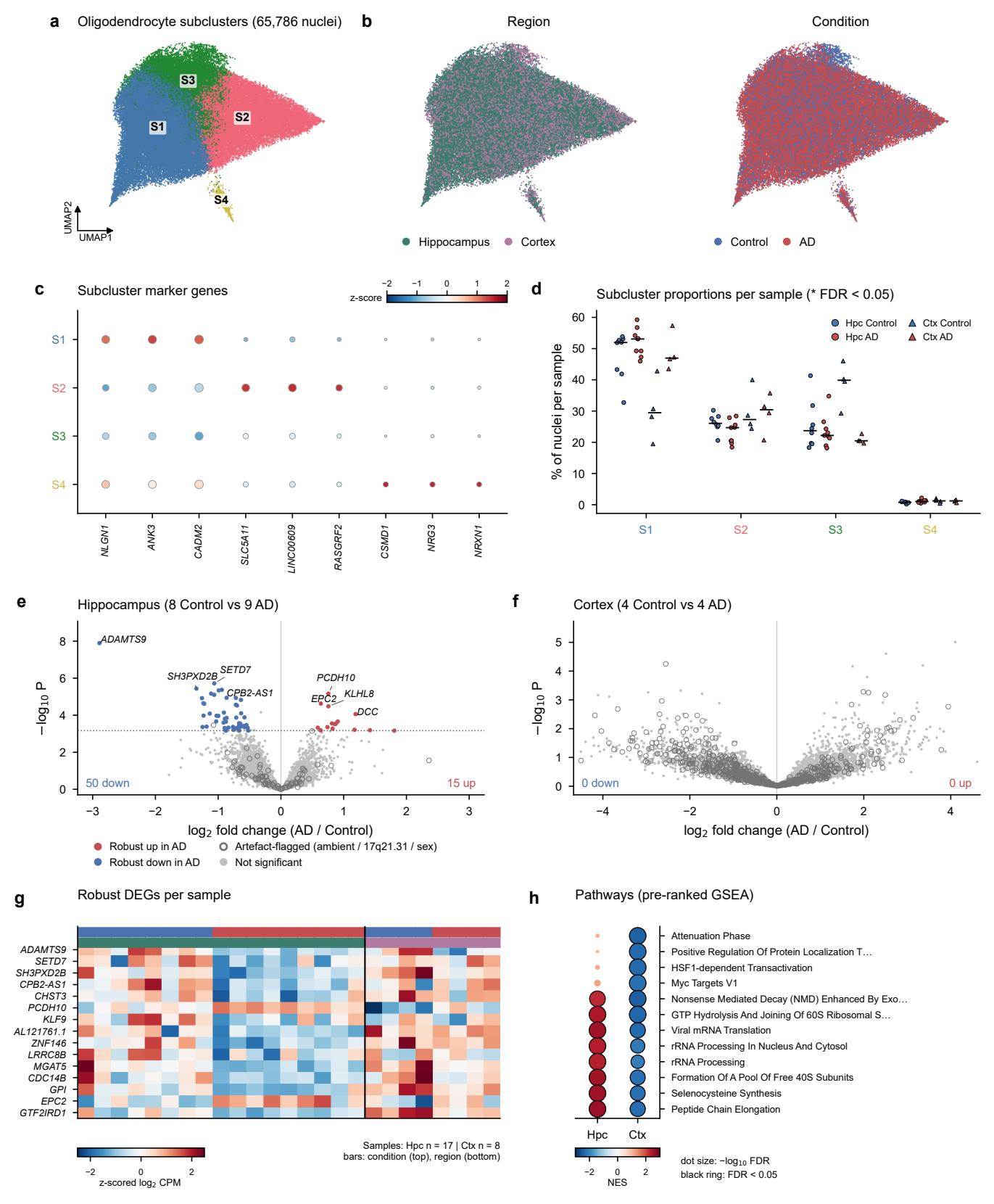

a
Oligodendrocyte subclusters (65,786 nuclei)
S1
S2
S3
S4
UMAP2
UMAP1
b
Region
Hippocampus
Cortex
Condition
Control
AD
c
Subcluster marker genes
z-score
−2 −1 0 1 2
S1
S2
S3
S4
NLGN1
ANK3
CADM2
SLC5A11
LINC00609
RASGRF2
CSMD1
NRG3
NRXN1
d
Subcluster proportions per sample (* FDR < 0.05)
% of nuclei per sample
Hpc Control
Hpc AD
Ctx Control
Ctx AD
S1
S2
S3
S4
e
Hippocampus (8 Control vs 9 AD)
−log10 P
ADAMTS9
SH3PXD2B
SETD7
CPB2-AS1
PCDH10
EPC2
KLHL8
DCC
50 down
15 up
log2 fold change (AD / Control)
Robust up in AD
Robust down in AD
Artefact-flagged (ambient / 17q21.31 / sex)
Not significant
f
Cortex (4 Control vs 4 AD)
−log10 P
0 down
0 up
log2 fold change (AD / Control)
g
Robust DEGs per sample
ADAMTS9
SETD7
SH3PXD2B
CPB2-AS1
CHST3
PCDH10
KLF9
AL121761.1
ZNF146
LRRC8B
MGAT5
CDC14B
GPI
EPC2
GTF2IRD1
z-scored log2 CPM
Samples: Hpc n = 17 | Ctx n = 8
bars: condition (top), region (bottom)
h
Pathways (pre-ranked GSEA)
Attenuation Phase
Positive Regulation Of Protein Localization T…
HSF1-dependent Transactivation
Myc Targets V1
Nonsense Mediated Decay (NMD) Enhanced By Exo…
GTP Hydrolysis And Joining Of 60S Ribosomal S…
Viral mRNA Translation
rRNA Processing In Nucleus And Cytosol
rRNA Processing
Formation Of A Pool Of Free 40S Subunits
Selenocysteine Synthesis
Peptide Chain Elongation
Hpc
Ctx
NES
dot size: −log10 FDR
black ring: FDR < 0.05